

VIT BHOPAL UNIVERSITY

CSE0001 – Digital Literacy

TASK 5: CYBERCRIME AWARENESS

Case Study & Prevention Guide

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PART A: CYBERCRIME CASE STUDY – PHISHING ATTACK

1. What is Phishing?

Phishing is a type of cybercrime in which attackers send fraudulent emails, messages, or links that impersonate trusted organizations – such as banks, universities, job portals, or government agencies – with the intention of tricking victims into revealing sensitive personal information such as passwords, OTPs, bank credentials, or personal identification details.

Type	Phishing Attack – Email-based Social Engineering
Target	College students actively seeking internships and jobs
Medium	Email / WhatsApp / Fake Websites

2. Realistic Case Study Scenario

The following is a realistic fictional scenario based on common phishing attack patterns reported in India. It describes how a college student fell victim to a phishing attack while seeking an internship opportunity.

SAMPLE PHISHING EMAIL (Reconstructed for Awareness)

From: hr.internship@techcompany-careers.in
To: student@vitbhopal.ac.in
Subject: URGENT: Internship Selection – Immediate Action Required

Dear Candidate,

Congratulations! You have been shortlisted for our exclusive paid internship program offering Rs. 15,000/month. Due to limited positions, we request you to complete your verification process IMMEDIATELY.

Kindly fill the form using the link below to confirm your slot:

[Click Here to Apply – Offer expires in 12 hours]

Failure to respond may result in cancellation of your selection.

*Regards,
HR Team – TechCompany Careers*

3. How the Attack Happened – Step by Step

Step 1	Attacker Prepares	The attacker creates a fake email address resembling a real company (techcompany-careers.in instead of the real domain). They design the email with professional branding, logos, and official-sounding language.
Step 2	Urgency is Created	The email creates extreme urgency – '12 hours to respond', 'limited slots', 'offer expires soon' – to prevent the victim from thinking critically or verifying the source.
Step 3	Victim Receives Email	Ravi, a first-year engineering student actively applying for internships, receives this email. The professional appearance and exciting opportunity cause him to act quickly without verification.
Step 4	Link is Clicked	Ravi clicks the link which redirects to a fake website designed to look identical to a real company portal. The URL is slightly altered: techcompany-careers.in instead of techcompany.com.
Step 5	Credentials Entered	The fake website asks for email ID, password, and additional personal details 'for verification purposes.' Ravi enters his information believing the process is legitimate.
Step 6	Data is Captured	The attacker immediately captures Ravi's credentials. Within hours, Ravi begins receiving alerts about suspicious login attempts, password reset notifications, and unauthorized account activity.
Step 7	Damage Occurs	Ravi's email account is compromised and used to send spam. His personal information is potentially sold or used for identity theft. He experiences financial anxiety and loss of trust in online opportunities.

4. Warning Signs Ravi Missed

- ■ *The sender's domain was 'techcompany-careers.in' – not the official company domain*
- ■ *Legitimate companies never ask for email passwords for 'verification'*
- ■ *The extreme urgency ('12 hours', 'immediate action') is a classic pressure tactic*
- ■ *No official company name, address, or verifiable contact details were provided*
- ■ *The link URL did not match any official company website*
- ■ *Genuine internship offers come through official portals or college placement cells*

5. Who is Usually Targeted?

College students in their first and second years are prime targets for phishing attacks because they are actively seeking internships and jobs, eager for opportunities, and may lack experience in recognizing fraudulent communications. Their eagerness to respond quickly to opportunities overrides their learned knowledge about cyber safety.

6. Consequences of the Attack

Identity Theft	Personal information submitted on the fake form can be used to impersonate the victim for financial fraud or account takeovers.
Account Compromise	Email, social media, and other accounts using the same password are immediately at risk of unauthorized access.
Financial Loss	If banking or UPI credentials were shared, direct financial theft can occur within minutes.
Mental Stress	Victims experience anxiety, embarrassment, and loss of trust in online platforms and opportunities.
Reputational Damage	Compromised accounts may be used to send spam or malicious content to the victim's contacts.

Source References

National Cyber Crime Portal – cybercrime.gov.in | CERT-In India Security Advisories – cert-in.org.in | Ministry of Home Affairs Cyber Safety Guidelines, Government of India

PART B: STAY SAFE ONLINE – PREVENTION CHECKLIST

The following 10-point checklist provides specific, actionable prevention tips designed for college students in India. Each tip addresses a real behavioral pattern exploited by cybercriminals. Two tips specifically address UPI and financial safety as required.

- ✓ Always verify the sender's email address domain carefully before clicking any link. Fake addresses often substitute letters: 'techcompany-careers.in' instead of 'techcompany.com'.
- ✓ Never click links in unsolicited emails or WhatsApp messages. Instead, go directly to the official website by typing the URL manually in your browser.
- ✓ Enable Two-Factor Authentication (2FA) on all important accounts including Gmail, GitHub, LinkedIn, and banking apps. This adds a critical second layer of security.
- ✓ Never share your OTP, password, or PIN with anyone – not even someone claiming to be from your bank, college, or a company. No legitimate organization ever asks for this.
- ✓ Use strong, unique passwords for each platform. A strong password contains a mix of uppercase letters, lowercase letters, numbers, and special symbols (e.g., An@kit2006!).
- ✓ Always check for HTTPS and a padlock icon in the browser address bar before entering any personal information on a website. HTTP without the 'S' is not secure.
- ✓ Keep your smartphone, apps, and browser updated regularly. Security updates patch known vulnerabilities that attackers actively exploit.
- ✓ [UPI Safety] Never scan a QR code sent by a stranger online. Unlike what many believe, scanning a QR code can DEBIT money from your account – not credit it.
- ✓ [UPI Safety] Your UPI PIN is NEVER required to RECEIVE money. If anyone asks for your UPI PIN to send you money, it is 100% a scam – disconnect immediately.
- ✓ If you suspect any cyber fraud or online scam, report it IMMEDIATELY at: Website: cybercrime.gov.in | National Helpline: 1930 (available 24x7, toll-free)

QUICK RULE TO REMEMBER

STOP – THINK – VERIFY

Before clicking any link, sharing any information, or responding to any urgent message online – STOP for 30 seconds, THINK about whether this seems suspicious or too good to be true, and VERIFY through official channels before proceeding.

REFLECTION

What Surprised Me Most

What surprised me most while researching this topic was how sophisticated and convincing modern phishing attacks have become. They no longer look obviously fake – they use real company logos, professionally written language, and create urgency that bypasses rational thinking even in people who are aware of cyber threats. The fact that Ravi, who had been taught about phishing in class, still fell for it demonstrates that awareness alone is not enough – behavior change is what truly protects us.

One Habit I Will Personally Change

The most important habit I will personally change as a result of completing this task is to always verify the sender's email domain carefully and check the URL of any website before entering credentials, especially for opportunities that seem urgent or too good to be true. I will also enable 2FA on all my important accounts immediately, and I will never enter my UPI PIN unless I am initiating a payment myself. These simple but deliberate habits can prevent the vast majority of cybercrime incidents.

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